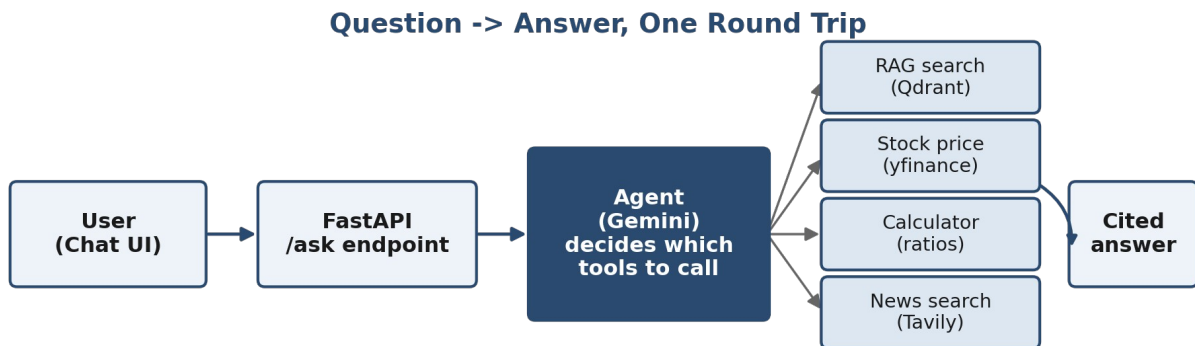


Finance AI Assistant

Project Documentation — Dataset to Deployment

Sri Sakticharan · September 2026



each tool returns raw data only — the agent writes the one final synthesized answer

A retrieval-augmented finance agent over real SEC filings, live prices, and news — built and deployed end to end.

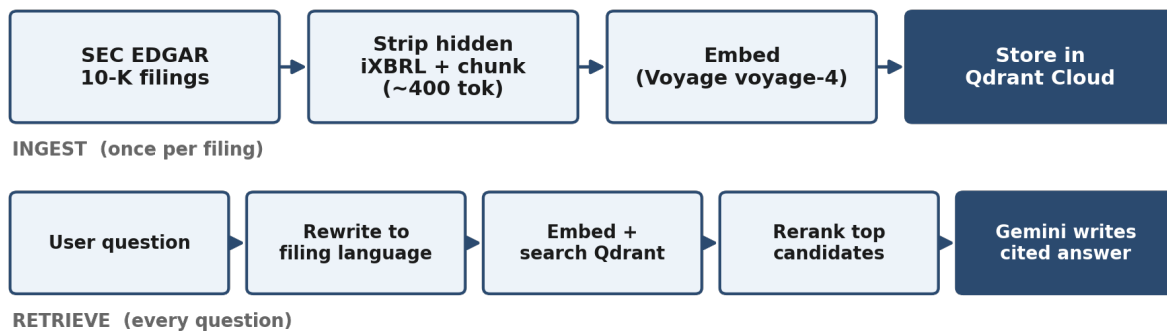
1. Overview

An AI agent that answers finance questions on Apple, Microsoft, and Nvidia using three real, live sources instead of model memory: their actual SEC 10-K filings (RAG), live stock prices, and news. The agent decides which tools it needs per question and writes one cited answer.

System at a Glance

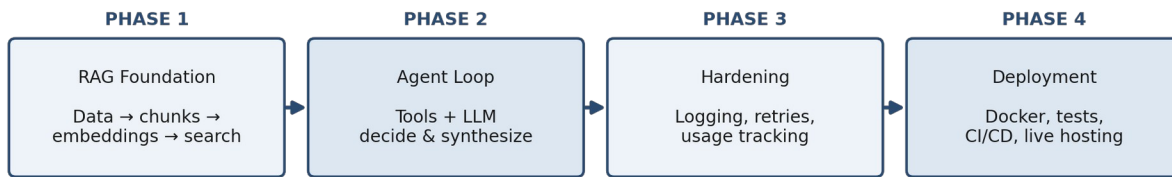
Layer	Technology
Data source	SEC EDGAR — real 10-K filings
Retrieval	Voyage AI embeddings + Qdrant vector DB
Agent / reasoning	Gemini (gemini-3.1-flash-lite)
Tools	RAG search, yfinance, calculator, Tavily news
Backend	FastAPI (Python)
Frontend	React + TanStack Start
Packaging	Docker + Docker Compose
CI/CD	GitHub Actions — test, build, deploy
Hosting	Render (API) + Vercel (UI) + Qdrant Cloud

2. How a Question Flows Through the System



Filings are chunked (~400 tokens, structure-aware) and embedded once. Every question is rewritten into filing-style language, embedded, searched, reranked, and answered — retrieval and generation are separate steps, not one blended call.

3. The Build, Phase by Phase



Phase 1 — RAG Foundation

- Pulled real 10-Ks directly from SEC EDGAR's free API — no paid wrapper.
- Bug: hidden iXBRL data leaked into extracted text; fixed by stripping hidden HTML nodes first.
- Recursive chunking (~400 tokens, 50-token overlap) + Voyage voyage-4 embeddings (the free-tier-eligible model).
- Self-hosted Qdrant via Docker — a deliberate learning choice over a managed DB.
- Hardest bug: retrieval confidently returned a wrong chunk. Fixed in five layers — company filtering, query rewriting, wider candidate pool, reranking, section-aware chunking. One sub-section edge case documented as a known limitation rather than over-engineered away.

Phase 2 — Agent Loop

- Wrapped RAG as one tool among four: RAG search, stock price, calculator, news search.
- Fixed truncated / report-style answers with a higher token limit and a synthesis-focused prompt.
- Tools return raw data only — one synthesis step keeps cross-tool reasoning intact.
- Switched Claude → Gemini to hit a hard \$0/month budget. Hit a real 429 rate-limit crash on the first model choice; fixed by picking a higher-quota free model, not by adding retries. Reversible in 2–3 files.

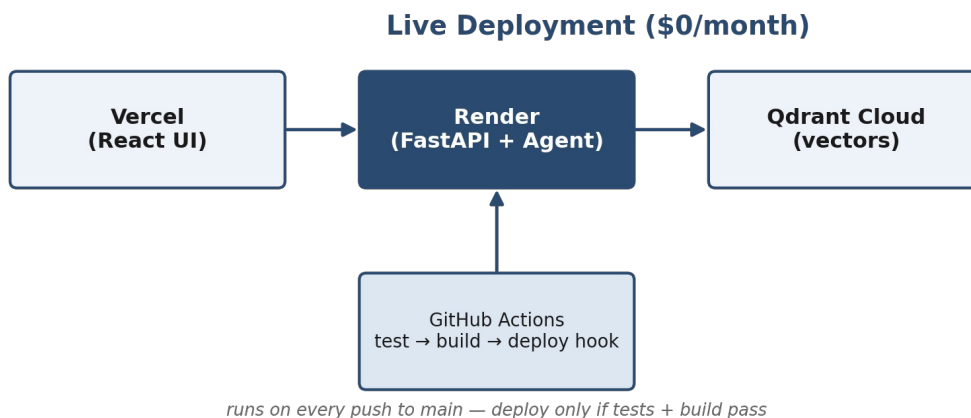
Phase 3 — Production Hardening

- One shared logger (console + file), wired into retrieval scoring and the agent's tool-calling loop.
- Centralized retry/backoff wrapper around every Gemini call (tenacity, exponential backoff).
- Lightweight per-day, per-provider usage tracker to watch free-tier limits before they're hit.

Phase 4 — Deployment

- Thin FastAPI layer (one endpoint, one line calling the existing agent) — validated via auto-generated docs.
- Docker caught two real bugs immediately: a hardcoded localhost that would've broken container-to-container networking, and missing dependencies only present locally, never in requirements.txt.
- 7 pytest tests on pure functions only — no paid/rate-limited API calls in CI.
- GitHub Actions: test → build → deploy, gated so a failing test blocks deployment via a private Render deploy hook.
- Bug found post-deploy: the agent fabricated unsourced facts (a fake CEO transition) because its system prompt never inherited the RAG prompt's "no outside knowledge" rule. Fixed and re-tested against the exact failure case.

4. Live Deployment



Render was chosen over Fly.io after research found Fly.io's no-card free tier ambiguous. Qdrant runs on Qdrant Cloud, not Render, since Render's free tier has no persistent disk — vector data would be lost on every idle restart.

Cost — \$0/month

Component	Provider / Tier
LLM	Gemini 3.1 flash-lite — free
Embeddings + rerank	Voyage AI — free 200M tokens
Vector DB	Qdrant Cloud — free tier
News + market data	Tavily + yfinance — free
Hosting	Render + Vercel — free tiers
CI/CD	GitHub Actions — free

5. Deliberately Not Built: Kubernetes

Kubernetes was the original final phase and was dropped on purpose. It solves problems this project doesn't have — auto-scaling and multi-node self-healing for a personal-demo traffic level. Local Minikube is free, so cost wasn't the real blocker: a local cluster would be a second, disconnected environment, unreachable from the internet and outside the real CI/CD pipeline. CI/CD, by contrast, was treated as a non-negotiable baseline regardless of scale.

6. Current State

- Live end to end: Vercel UI → Render API → agent → Qdrant Cloud / Gemini / Voyage / Tavily / yfinance.
- Every push to main tests, builds, and deploys automatically — only if tests and build pass.
- Known, accepted limits: Render cold-start after idle, Qdrant Cloud sleep after a week idle, sub-section chunking gap, tests scoped to pure functions only.

Full decision-by-decision detail, alternatives considered, and interview-ready answers live in the companion Decision Log.